Ultrasound in Medicine & Biology 51 (2025) 638–649

Contents lists available at ScienceDirect

Ultrasound in Medicine & Biology

journal homepage: www.elsevier.com/locate/ultrasmedbio

Original Contribution

Regional Image Quality Scoring for 2-D Echocardiography Using Deep Learning

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ARTICLE INFO

Keywords:
Cardiac segmentation
Ultrasound
Image quality
Coherence
Signal-to-noise ratio

ABSTRACT

Objective: To develop and compare methods to automatically estimate regional ultrasound image quality for echocardiography separate from view correctness.
Methods: Three methods for estimating image quality were developed: (i) classic pixel-based metric: the generalized contrast-to-noise ratio (gCNR), computed on myocardial segments (region of interest) and left ventricle lumen (background), extracted by a U-Net segmentation model; (ii) local image coherence: the average local coherence as predicted by a U-Net model that predicts image coherence from B-mode ultrasound images at the pixel level; (iii) deep convolutional network: an end-to-end deep-learning model that predicts the quality of each region in the image directly. These methods were evaluated against manual regional quality annotations provided by three experienced cardiologists.
Results: The results indicated poor performance of the gCNR metric, with Spearman correlation to annotations of $\rho = 0.24$. The end-to-end learning model obtained the best result, $\rho = 0.69$, comparable to the inter-observer correlation, $\rho = 0.63$. Finally, the coherence-based method, with $\rho = 0.58$, out-performed the classical metrics and was more generic than the end-to-end approach.
Conclusion: The deep convolutional network provided the most accurate regional quality prediction, while the coherence-based method offered a more generalizable solution. gCNR showed limited effectiveness in this study. The image quality prediction tool is available as an open-source Python library at https://github.com/GillesVanDeVyver/arqee.

Introduction

Image quality is one of the main challenges in ultrasound imaging and can differ significantly between patients and imaging equipment. Many factors influence image quality in echocardiography such as the ultrasound scanner, the patient and the probe. Several quantitative measurements using the images are performed. However, this requires image quality good enough for the given measurement. Different measurements have different image quality requirements; *e.g.*, left ventricular (LV) volume, ejection fraction (EF) and strain measurements require good image quality across the entire myocardium. On the other hand, mitral annular plane systolic excursion only requires good image quality in the annulus. We also believe it is important to estimate regional image quality for each frame. For instance, end-diastole and end-systole frames are used for EF measurements, thus good image quality is required for both those frames. For measurements such as strain, which uses frame-by-frame tracking, the image quality of each segment of every frame is important as low quality in some frames may ruin the tracking in that segment. Good image quality should generally provide measurement values with low uncertainty. Estimating image quality can be useful in the following ways:

- To guide operators to achieve as good image quality as possible while scanning.
- To automatically select the best images and recordings, and the best cardiac cycles to use for a given measurement.
- As quality assurance, *e.g.*, to warn the user when an image is not good enough for a measurement, and to automatically approve/disapprove individual myocardial segments based on image quality.

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https://doi.org/10.1016/j.ultrasmedbio.2024.12.008
Received 1 August 2024; Revised 16 December 2024; Accepted 18 December 2024

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- In data mining projects, to exclude cases with insufficient quality for reliable measurements.

Image quality is subjective and can vary depending on the application or the specific measurement being performed. What qualifies as "good" or "acceptable" depends on the context. In this study, we used the definitions outlined in Table 1.

We distinguished between two types of quality of ultrasound images: view quality/correctness and image quality. In this work we focused on image quality specifically. For view correctness, previous work demonstrated that 3-D ultrasound can serve as training data to automatically identify the transducer rotation and tilt in relation to the desired standard view and can guide the user to the correct position [1–3]. For image quality, the classic ultrasound signal-processing metrics include contrast ratio (CR) [4], contrast-to-noise ratio (CNR) [5] and generalized CNR (gCNR) [6]. These three metrics need a region of interest (ROI) and a background region to compare against. More recently, global image coherence (GIC) [7] was proposed as a general quality metric that does not require the selection of these two regions. Image coherence measures how well the signals of the transducer elements align after delay compensation, with more alignment corresponding to clearer and sharper images. From the above-mentioned methods, only GIC can be used directly and automatically for measuring image quality separately as it does not require selecting an ROI and noise region. However, this approach requires channel data, which are not readily available in practice and do not give regional metrics.

Several automatic methods for measuring ultrasound image quality have been published applicable to cardiac imaging. Abdi et al. [8] used a recurrent neural network to predict the global quality of cardiac cine loops. Criteria for quality assessment took both image quality and view correctness into account. In subsequent studies [9,10], they used an architecture that performed both view classification and global quality prediction simultaneously. The image quality metric is a global criterion based on the manual judgment of the clarity of the blood-tissue interface. Labs et al. [11] used a multi-stream neural network architecture where each stream took in a sequence of frames and predicted a specific quality criterion. The criteria are global and take both view correctness and image quality into account. Karamalis et al. [12] detected attenuated shadow regions with random walks resulting in a pixel-level confidence map. Unlike the other methods above, this method was not based on deep learning. It provided a local, pixel-level metric, but only measured the visibility of regions and not the quality of their content.

All of the automatic methods mentioned above have the limitation that they only provide a global image quality evaluation and/or do not assess image quality separate from view correctness. The novelty of this work is automatic quality estimation on a regional level. To the best of our knowledge, this is the first study to propose a method for automatic regional image quality assessment in echocardiography. Our method focuses specifically on quantifying image quality within different myocardial segments and in the annulus points, which can provide more granular and precise insights into the suitability of specific regions for various measurements.

Table 1
Definitions of image quality labels for annotating cardiac regions of interest

Image quality label	Definition	Quality score
Not visible	Less than 50% of what is inside the sector is visible	1
Poor	Low signal-to-noise ratio and/or smearing/low resolution, but can still see at least 50% of the myocardium/annulus. Diffuse border between annulus and cavity	2
Ok	Medium signal-to-noise ratio and/or some smearing/lower resolution, can see at least 70% of the myocardium/annulus	3
Good	Good signal-to-noise ratio and clear border in at least 80% of the myocardium/annulus. Very little smearing/good resolution. Clear border between annulus and cavity	4
Excellent	Within the best 10% of what you have seen on this system	5

Only the signal quality of the region that was inside the ultrasound sector was analyzed. If more than 50% was outside the sector, the region was treated as out-of-sector and excluded from analysis.

Methods

In this work, we developed and compared three fully automatic methods to assess regional image quality in cardiac ultrasound separate from view correctness:

- Classical ultrasound image quality metrics, such as CR and CNR, applied in cardiac regions automatically extracted using deep-learning segmentation.
- Deep learning-predicted ultrasound coherence, which is a measure of how coherent a signal is received by the transducer elements, together with deep-learning segmentation.
- End-to-end prediction of regional image quality.

The rest of this section presents the datasets used to develop these methods and then presents each of the three methods.

Datasets

VLCD. The Very Large Cardiac Channel Data Database (VLCD) consists of channel data from 33,280 frames from 538 recordings of 106 study participants. It contains parasternal short axis (PSAX), parasternal long axis (PLAX), apical long axis (ALAX), apical two-chamber (A2C) and apical four-chamber (A4C) views. We split the VLCD dataset at the study participant level into train, validation and test sets, allocating 70%, 15% and 15%, respectively.

HUNT4. The Nord-Trøndelag Health Study dataset (HUNT4Echo) is a clinical dataset including, among others, PSAX, PLAX, ALAX, A2C and A4C views acquired using a GE Vivid E95 scanner and GE M4S probe. Each recording contains three cardiac cycles. We used two subsets of the HUNT4Echo dataset:

- Segmentation annotation dataset: a fraction of 311 study participant exams, the segmentation annotation set [13] contains single-frame segmentation annotations in both end diastole and end systole as pixel-wise labels of the LV, left atrium (LA) and myocardium (MYO) in ALAX, A2C and A4C views.
- Regional image quality dataset: for this work, we created an additional dataset of image quality labels. Local image quality labels were manual annotations that assessed the image quality of the cardiac ROIs on a subset of the HUNT4 dataset in ALAX, A2C and A4C views.

Regional image quality annotation on HUNT4

An annotation tool was developed specifically for this project using open-source Annotation Web software (https://github.com/smistad/annotationweb) [14], which was developed to enable clinicians to annotate regional image quality as efficiently and accurately as possible. The tool is freely available and can be adapted to other image quality projects. Table 1 defines the quality levels used in this work. Three cardiologists, each of whom performed more than 10,000 echocardiographic examinations and were European Association of Cardiovascular Imaging certified in transthoracic echocardiography, performed the quality annotations. The cardiologists used the following protocol:

1. Annotate the end-diastole and end-systole frame of each recording, and optionally other frames if the image quality changes significantly during the recording.
2. If the majority of the cardiac ROIs is out of sector, label it as out of sector. Otherwise, label the part of the region that is inside the sector according to the definitions in Table 1. Out-of-sector regions were ignored in the remainder of this work.

For the first round of annotations, each of the three clinicians annotated the same 10 frames from 5 recordings of 2 study participants. This dataset was used to calculate the inter-observer variability. For the second round of annotations, the three clinicians collectively annotated 458 frames from 158 recordings of 65 study participants. The annotations from the second round formed the regional image quality dataset. This dataset was split randomly at study participant level into train, validation and test sets, allocating 70%, 15% and 15% of the data to each set, respectively. Table 2 shows the consistency of each split across the ALAX, A2C and A4C views.

Regional image quality estimation

Classical ultrasound image quality metrics. For classical image quality metrics, deep-learning segmentation was used to extract the annulus regions and each of the myocardial segments as ROIs, with the LV as the background region. Appendix A gives more details on the procedure for dividing the segmentation into regions. The four classical ultrasound image quality metrics below were tested. We applied histogram matching [15,16] to a Gaussian distribution ($\mu = 127, \sigma = 32$) for the B-mode grayscale images before calculating pixel-based quality metrics.

Table 2
Number of study participants, recordings and frames in each set of the regional image quality dataset for each view

Set	Number of ALAX study participants/ recordings/frames	Number of A2C study participants/ recordings/frames	Number of A4C study participants/ recordings/frames
Train	31/31/99	42/43/120	33/34/102
Validation	8/8/20	10/10/31	8/8/27
Test	7/7/18	9/9/22	8/8/19

Participants may appear in multiple views, but each was in only one of the train, validation or test sets. ALAX, apical long axis; A2C, apical two-chamber; A4C, apical four-chamber.

- Pixel intensity is the average pixel intensity value in each region.
- Contrast ratio (CR) [4] was defined as (eqn [1]):

$$CR = \frac{\mu_{\text{segment}}}{\mu_{\text{LV}}} \tag{1}$$

where μ_{segment} is the average intensity in each region and σ_{LV} is the average intensity inside the LV lumen.

- Contrast-to-noise ratio (CNR) [5] was defined as (eqn [2]):

$$CNR = \frac{\mu_{\text{segment}} - \mu_{\text{LV}}}{\sqrt{\sigma^2_{\text{segment}} + \sigma^2_{\text{LV}}}} \tag{2}$$

where μ_{segment} is the standard deviation in each region and σ_{LV} is the standard deviation inside the LV lumen.

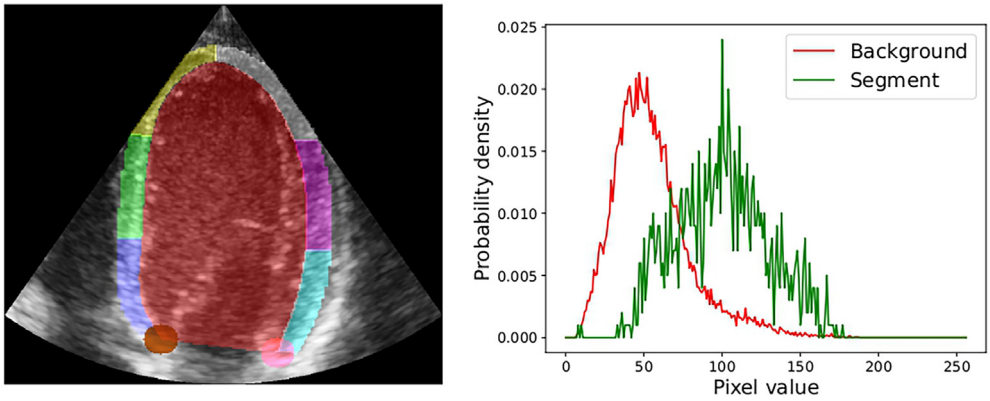
- Generalized CNR (gCNR) [6] is defined as the maximum performance that can be expected from a hypothetical pixel classifier based on intensity using a set of optimal thresholds. It was calculated as follows (eqn [3]):

$$gCNR = 1 - \frac{1}{2}\sum_{i=0}^{MAX_i} \min\left\{p_{\text{segment}}(i), p_{\text{LV}}(i)\right\} \tag{3}$$

where $p_{\text{segment}}(x)$ is the probability density function of the pixel intensities inside the region the gCNR is calculated for, $p_{\text{LV}}(x)$ is the probability density function of the pixel intensities inside the LV lumen, and *MAX* is the maximum possible pixel intensity. Figure 1 shows an example of the probability density functions for one of the regions as ROI and the LV lumen as background.

Local, deep-learning predicted image coherence as quality metric

We used the VLCD dataset to calculate the coherence factor [17] for each pixel in the ultrasound image. This factor is the ratio between the amplitude of the sum of the received signals to the sum of the amplitudes of those signals, where S_i is the delayed signal for the i-th transducer element. This is equivalent to taking the coherent sum of the signal and dividing it by the incoherent sum of each signal. Thus, the coherence factor measures how well the complex signals of all transducer elements align. The remainder of the signal-processing chain comprised native processing of the GE HealthCare Vivid E95 system but



(a) Segmentation (b) Probability density functions

Figure 1. Calculating the generalized contrast-to-noise ratio (gCNR) for a region of the myocardium. The left side of the figure shows segmentation with the myocardium and annulus points divided into regions. The right side shows the probability density functions of the segment and background pixels used to calculate the gCNR. In this example, we use the mid region on the left side as the region of interest and the full left ventricular lumen as background. These respectively correspond to the green and red masks in the left part of the figure.

without log compression. This signal-processing chain is described in more detail by Gundersen et al. [18]. The result was a coherence image with the same dimension as the B-mode image. The final preprocessing step applied gamma normalization with $\gamma = 0.5$ on the coherence images (eqn [4]):

$$t_{i,normalized} = t_i^{\gamma} \tag{4}$$

where t_i are the pixels of the target coherence image. Corresponding B-mode images were generated from channel data using the same native signal-processing pipeline.

The HUNT4 dataset, like most ultrasound datasets, does not include channel data. Therefore, VLCD was used to train an image-to-image network that took as input the grayscale B-mode image and predicted the coherence image, which was then used to calculate local image coherence. We used a lightweight U-Net architecture inspired by the U-Net 1 architecture in [19], with the characteristics listed in Table 3. As coherence is related to image quality, we only applied augmentations that did not influence the quality of the image. Furthermore, coherence should be invariant to different gain settings, so we additionally augmented with brightness adjustments on the B-mode image while keeping the target coherence image unchanged. During training and validation of the coherence prediction model, we sampled a random frame from each recording in the training and validation sets, respectively, during each epoch. During testing, we used all frames in the test set. The local image coherence quality metric of a region is the average pixel value of all pixels corresponding to the region in the coherence image, which is the same as the pixel intensity metric above but applied to the coherence image instead of the B-mode image.

End-to-end deep-learning quality prediction

The end-to-end learning approach trains a convolutional neural network on the regional image quality dataset to predict the quality of each region directly. The network predicts the image quality labels of all regions simultaneously, as illustrated in Figure B1a. Regarding architecture, we started with MobileNetV2 [20] and replaced the final dense layer with a dense layer with eight outputs, one for each region. MobileNetV2 weights were initialized using a model pre-trained on ImageNet [21], with all weights set as trainable during training. We treated the problem as a regression task where the model predicted a score for each segment. Table 1 shows the correspondence between quality scores and annotation labels. The loss function was the sum of the mean squared errors of each output and the model with the best validation loss was selected. Table 4 lists the remaining configuration details used during training and Appendix B describes the ablation study conducted to justify this setup.

Table 3
Characteristics of the coherence prediction network

Input size	256 × 256
Number of channels	16 ↓ 128 ↑ 16
Number of output channels	1
Lowest resolution	4 × 4
Up-sampling scheme	2 × 2 repeats
Normalization scheme	BatchNorm
Batch size	32
Optimizer	Adam
Initial learning rate	1e-2
Scheduler	None
Loss function	Negative structural similarity index measure (SSIM) [1]
Inter-layer activation	Gaussian error linear unit (GELU) [2]
Final layer activation	Sigmoid
Epochs	500
Augmentations	Rotations (-45° ≤ angle ≤ 45°), horizontal mirroring, gamma correction ($0.75 \leq \gamma \leq 1.25$, only on B-mode), scaling ($0.75 \leq$ magnification ≤ 1.25), contrast/brightness adjustments ($0.2 \leq$ gain ≤ 2, $0 \leq$ bias ≤ 128, only on B-mode). Each augmentation is applied individually with 0.5 probability.

The general architecture was U-Net. The 'number of channels' row indicates the number of channels at the first, bottom and last convolution of the U-Net, respectively.

Experimental setup

Evaluation of coherence prediction from B-mode

We used the structural similarity index [22], peak signal-to-noise ratio (the maximum pixel value for coherence images is 1) and relative pixel error (RPE) to evaluate the coherence image prediction network. We defined the RPE as (eqn [5]):

$$RPE = \frac{\left| t_i - p_i \right|}{max(t_i, \varepsilon)} \tag{5}$$

where t_i are the pixel values of the target coherence image, p_i the pixel values of the predicted coherence image and $\varepsilon = 1e-4$.

Evaluation of quality metrics

The correlation and accuracy of each quality metric were measured by comparing them to the expert annotations on the test set of the regional image quality dataset. For the classic image quality methods and regional coherence method, linear regression models were used to map the quality metric values to image quality labels. Training and validation sets were used together to fit the linear regression model and evaluate it on the test set.

Comparison with inter-observer variability

We compared the end-to-end local image coherence and gCNR-based models with inter-observer variability on the data obtained in the first round of annotation. Inter-observer variability was calculated from the aggregate of the three unique pairwise score errors between each of the three annotators (eqn [6]):

$$e_{\text{inter-observer}} = e_{12} \cup e_{23} \cup e_{13} \tag{6}$$

where e_{ij} is the score difference between operator i and j. The error metrics of the automatic methods were calculated from the aggregate of the pairwise score errors between the output of the method and each of the three annotators (eqn [7]):

$$e_M = e_{1M} \cup e_{2M} \cup e_{3M} \tag{7}$$

where e_{iM} is the score difference between operator i and method M.

Table 4
Training configuration of the end-to-end learning model

Input size	256 × 256
Batch size	16
Optimizer	Adam
Initial learning rate	1e-4
Scheduler	None
Loss function regression	Mean squared error
Loss function classification	Squared earth mover's distance [3]
Epochs	500
Augmentations	Rotations (-30° ≤ angle ≤ 30°), horizontal mirroring (also mirror labels), gamma correction ($0.9 \leq 7 \leq 1.1$) and scaling ($0.85 \leq$ magnification ≤ 1.15). Each augmentation is applied individually with 0.5 probability

The final model uses regression, as motivated by the ablation study in Appendix B.

Table 5
Average metric values of the coherence prediction network on the test set of the Very Large Cardiac Channel Data Database

SSIM [1]	0.994 ± 0.003
Relative pixel error	5.6% ± 0.89%
Peak signal-to-noise ratio	35.99 ± 2.00 dB

SSIM, negative structural similarity index measure.

Relation to variability in clinical measurements

This experiment evaluated whether there is a relation between predicted quality and agreement between different methods for clinical measurements. We hypothesized that with lower image quality the variability, and thus uncertainty, of measurements between methods and experts would increase. More specifically, this analysis compared peak global longitudinal strain (GLS) and EF measurements obtained either fully automatically with AI tools or manually by using EchoPAC (GE Vingmed ultrasound AS, Norway) software on HUNT4 [13,23]. For AI estimations of GLS and EF, the respective deep-learning methods proposed by Stvik et al. [24] and Smistad et al. [25] were used. Study participants used for model development in HUNT4 were excluded from the analysis. For GLS, the predicted quality was the average quality of all segments over the full recording. For EF, the predicted quality was the average quality of all segments in the end-diastole and end-systole frames of all cycles in the recording.

Evaluation on CAMUS

To test the generalizability of the end-to-end model, we applied the model to each recording in the public Cardiac Acquisitions for Multi-structure Ultrasound Segmentation (CAMUS) dataset [26] and compared the predictions with the reference quality labels. The predicted quality was the average quality of all segments over the full recording.

Results

Results of coherence prediction from B-mode

Table 5 summarizes the average metric values on the test set. Figure 2 shows an example of the best, median and worst-case predictions according to the RPE. Figure 3 illustrates how the predicted coherence images were almost independent of the brightness/gain and contrast/dynamic range of the input B-mode images. The main finding was that the difference between the estimated and ground truth coherence images was small and thus the predicted coherence images could be used to obtain coherence-based quality metrics for B-mode, for which the channel data are not available.

Results of quality metrics

Table 6 summarizes the results of quality metric evaluation. Figure 4 shows box plots of the quality metrics per image quality label for the end-to-end, coherence, gCNR and intensity models. Figure 5 shows examples of B-mode images of varying quality together with labels from the annotators and automatic quality metrics. The main finding was that the end-to-end model performed the best, followed by the local image coherence metric. The classical ultrasound image quality metrics performed poorly.

Results of comparison to inter-observer variability

Figure 6 shows a bar plot comparing the automatic methods with inter-observer variability and Table 7 lists the corresponding average metric values. Using the Wilcoxon signed rank test [27] and a significance level of $p = 0.05$, we found that the difference in mean absolute error (MAE) between each of the methods was statistically significant. The difference between the inter-observer MAE and the MAE of each of the methods was also statistically significant, *i.e.*, the inter-observer

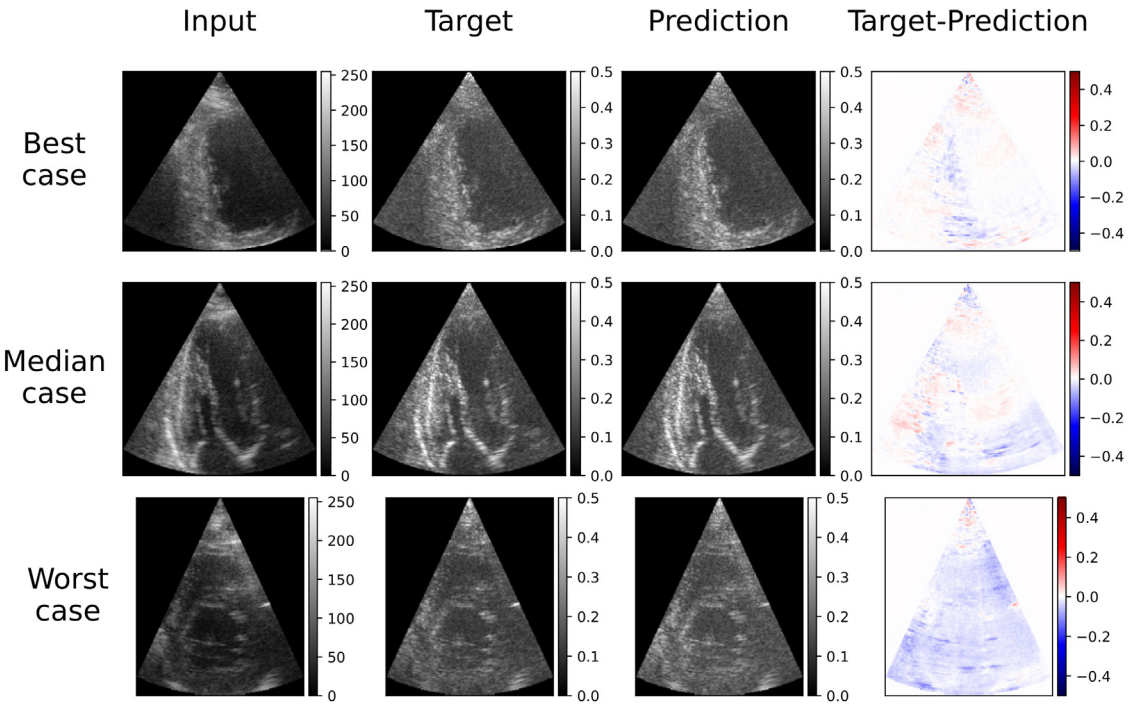


Figure 2. Best, median, and worst case of the image-to-image coherence prediction task with relative pixel errors of 4.0, 5.4, and 9.1% respectively. The first column shows the input B-mode image. The second column shows the coherence image as predicted by the image-to-image network. The third column shows the ground truth coherence image as calculated from the channel data. Finally, the right-most column shows the color-coded difference of the target minus the predicted image. The images originate from the Very Large Cardiac Channel Data Database dataset, which is only used to train the coherence network on the pixel level. Thus, the correctness of the view and its alignment are not crucial in this context.

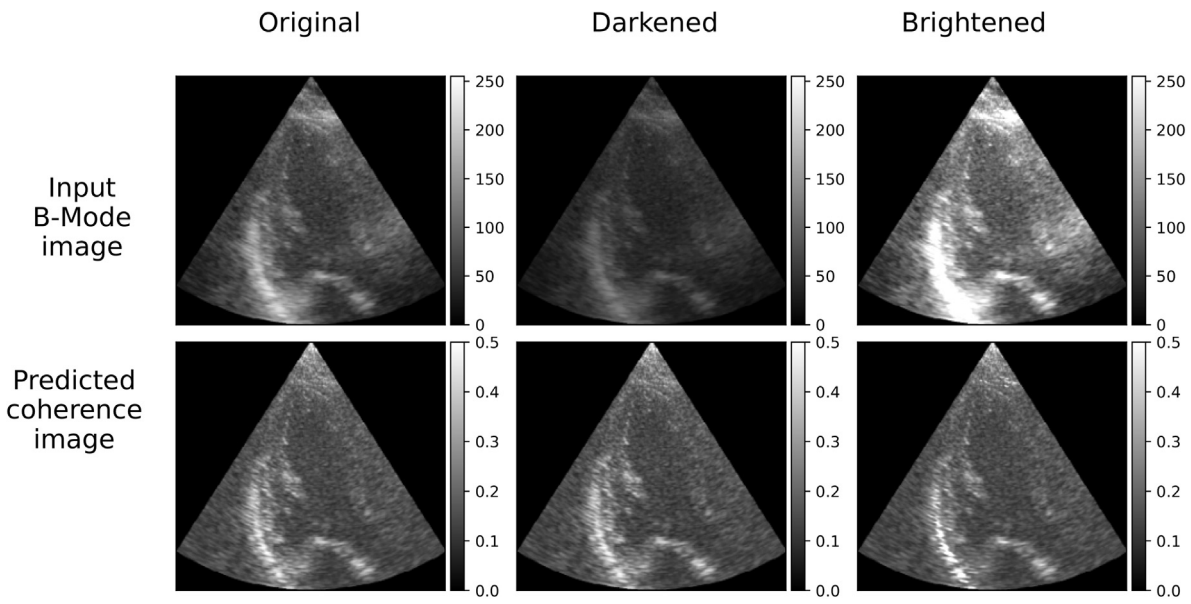


Figure 3. Effect of brightness on coherence prediction. The first row shows a B-mode image from the regional image quality dataset, brightened and darkened with gamma correction ($\gamma = 0.9$ and 1.1, respectively). The second row shows the predicted coherence images generated by giving the corresponding input from the first row to the network. The predicted coherence is unaffected by adjustments in brightness, apart from the saturation effect in the brightened image reducing the information in the input, as can be seen in the basal part of the inferolateral wall.

Table 6
Comparison of quality metrics on the test set of the regional image quality dataset, *i.e.*, the second round of annotations

Quality metric	Spearman correlation	Mean absolute error	Accuracy
Pixel intensity	0.30	1.31 ± 0.91	20%
CR	0.28	1.32 ± 0.92	22%
CNR	0.25	1.38 ± 0.96	20%
gCNR	0.20	1.50 ± 0.96	15%
Local image coherence	0.48	1.07 ± 0.80	29%
End-to-end learning	0.65	0.59 ± 0.45	48%

Accuracy is defined as the ratio of agreement between the automatic measurement, rounded to the nearest quality category, and the annotation on the segment level. CNR, contrast-to-noise ratio; CR, contrast ratio; gCNR, generalized contrast-to-noise ratio.

MAE was higher than the MAE of the end-to-end model and lower than the MAE of the other two models.

Results of relation to variability in clinical measurements

Figure 7 shows box plots visualizing the agreement between the measurements obtained automatically and with EchoPAC for each predicted quality category. The standard deviations in these plots represent how well the AI estimates agreed with the manual references. The main finding was that the limits of agreement were narrower for higher qualities.

Results of evaluation on CAMUS

Figure 8 shows the box plots of the average qualities over all frames as predicted by the end-to-end model for each quality category in CAMUS. Although the image quality categories in CAMUS had different definitions compared with those in this study, better quality on one scale should on average also relate to better quality on the other scale. Figure 8 confirms this is indeed the case. The difference in average predicted quality between the different quality categories in CAMUS was statistically significant using the independent two-sample t-test and a significance level of $p = 0.05$.

Discussion

Challenges and considerations

Assessing image quality based on human perception is inherently a subjective task, even when supported by clear definitions of image quality categories. This creates challenges for training and evaluation as there is no ground truth as the reference labels are themselves a subjective estimation. Therefore, it is not realistic to expect automatic models to agree with reference labels as well as on tasks with well-defined ground truth labels. This, together with a rather fine scale of image quality categories, explains the low accuracies seen in Tables 6 and 7. Figure 6 shows how the end-to-end model had on average less error than the annotators between each other. This indicates that the model learned to produce quality labels that struck a middle ground between the subjective assessments of the annotators.

The end-to-end learning model over-estimated low-quality regions and under-estimated high-quality regions, as can be seen in Figure 4a. This means that the model could only explain a limited amount of variability in the image quality labels and was a result of minimizing the mean squared error while dealing with subjective and thus noisy reference values with fixed boundaries. We were able to eliminate this effect by fitting a linear model on the validation set that mapped predicted image quality to image quality labels and applying it when doing inference on the test set. This increased mean squared error but gave more uniform performance over the image quality labels.

One reason for the weak correlation between annotations and pixel-based methods was the rough selection of ROI and background region. Figure 4 shows how the average metrics of the classic pixel-based and coherence metrics increased for each quality label until the good label, and then drop again for excellent. This was because on the one hand in these high-quality images, the blood speckles were visible inside the LV lumen, which was used as background region, and on the other hand the myocardium tissue, which was used as ROI, was less blurred resulting in a smaller spread of pixels with high intensity. This could be seen for the anterolateral wall and apex in the right-most column of Figure 5. One possibility to only select regions belonging to the tissue was to perform automatic pixel selection methods such as Otsu thresholding [28] or percentile filters, but in our experiments this reduced performance further.

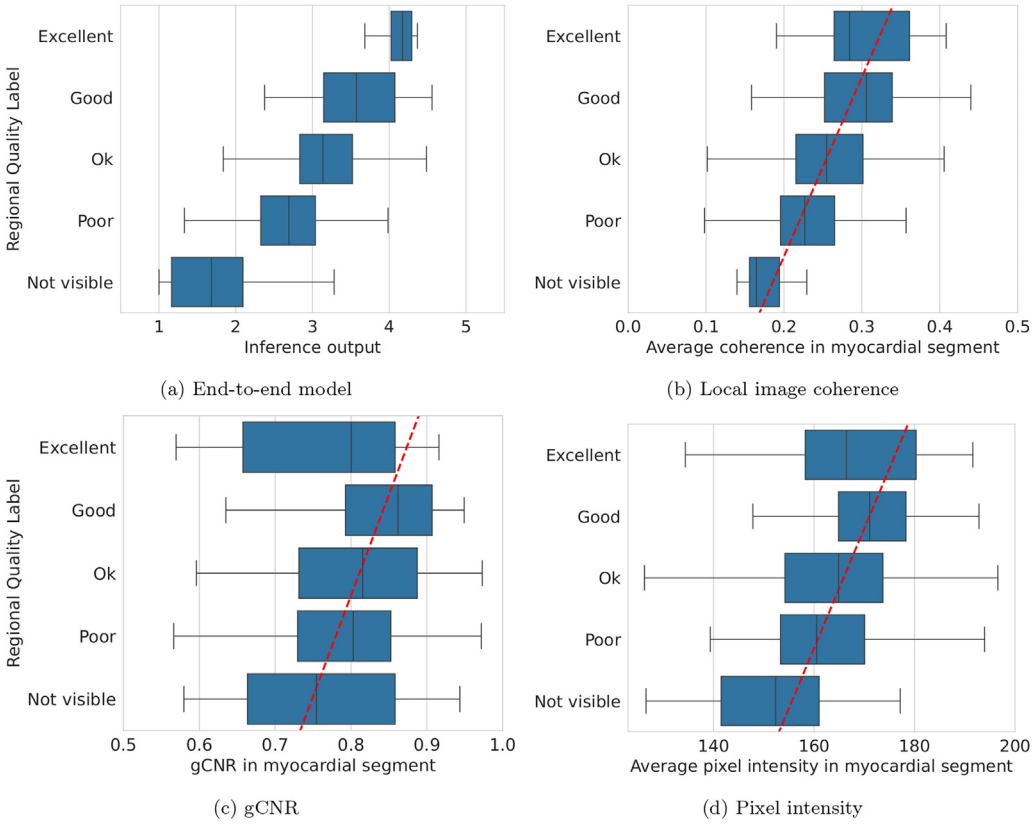


Figure 4. Box plots of quality metrics versus regional quality labels on the test set of the regional image quality dataset. Predictions of the end-to-end model have the strongest correlation to the quality labels. The dotted line represents the linear regression model that maps the quality metrics to quality labels. The inference output of the end-to-end model can be used directly without an additional linear model. gCNR, generalized contrast-to-noise ratio.

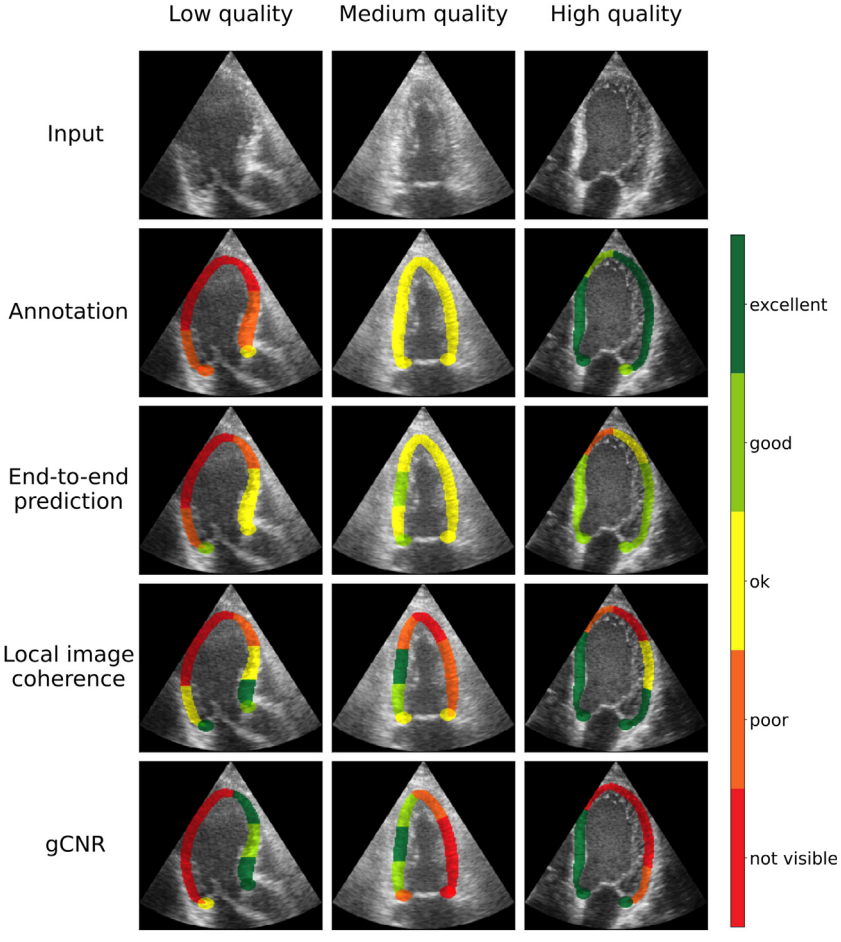


Figure 5. Example cases of annotations and automatically predicted regional quality from the test set. The visualization uses the regional quality metrics to color-code the output of divided segmentation output. The end-to-end model predicts the regional qualities directly from the B-mode without using the segmentation output. The local image coherence metric uses the segmentation output to select the region of interest (ROI). The gCNR metric uses the segmentation output to select the ROI and background region. The background region, *i.e.*, the left ventricular lumen, is not shown in the image. gCNR, generalized contrast-to-noise ratio.

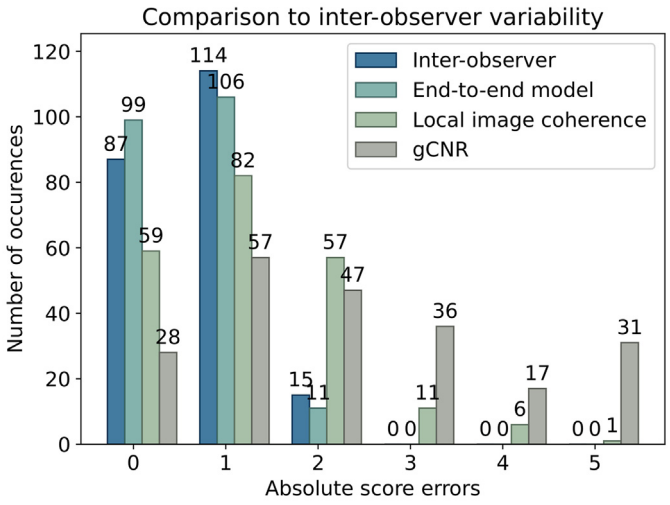


Figure 6. Bar plot comparing inter-observer variability with automatic methods. A method with lower variability will have the most occurrences with low score errors. Here we can observe that the variability of the end-to-end model is on par with the inter-observer variability, while the two other methods are not. gCNR, generalized contrast-to-noise ratio.

Table 7
Comparison of automatic methods to inter-observer variability on the first round of annotations

Method	Spearman correlation	Mean absolute error	Accuracy
End-to-end model	0.69	0.59 ± 0.59	46%
Local image coherence	0.58	1.19 ± 1.01	27%
gCNR	0.24	2.23 ± 1.58	13%
Inter-observer	0.63	0.67 ± 0.60	40%

For these data, there were three annotation values for each quality label. Each of the automatic methods was compared against each of the three annotators and the results averaged. For inter-observer variability, the labels were compared pairwise between each of the three annotators. gCNR, generalized contrast-to-noise ratio.

It could be argued that classic metrics such as (g)CNR measure something conceptually different to the qualitative assessment of clinicians. This work proposed a method to automate the extraction of these classic metrics and studied how well these aligned with the subjective ratings

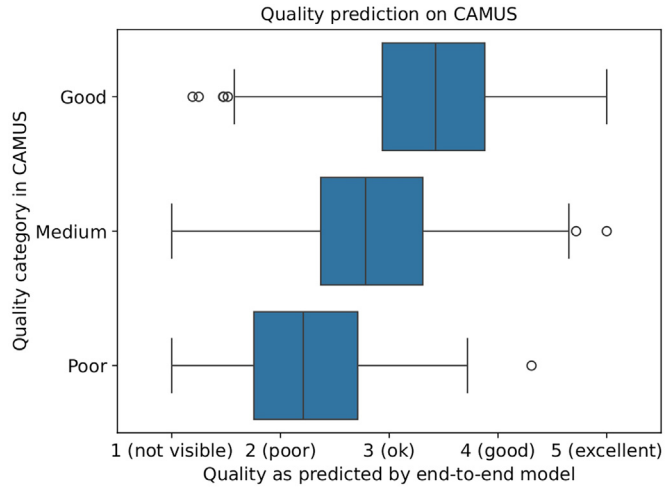
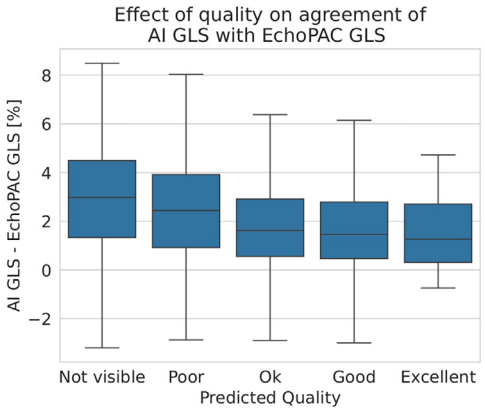


Figure 8. Box plots of the predicted quality scores for each quality category in the CAMUS dataset, based on the end-to-end models predictions.

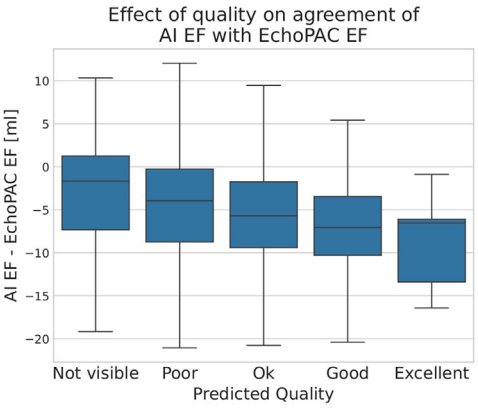
of clinicians. The weak correlation does not, however, necessarily mean that these metrics are inferior. Instead, the correct approach to measure image quality depends on the application, and we demonstrated that the classical metrics did not correlate well with the qualitative labels of the cardiologists in echocardiography.

Design choices

The different methods in this study had a trade-off between accuracy and versatility. The default end-to-end network gave the best results but required specific image quality labels for the task. Next, the coherence-based method was more generic and could potentially be applied more generally without the need for view-specific image quality annotations. Rindal et al. [7] showed that the GIC was not significantly different between apical views but higher for apical views than parasternal views. Thus, while a single image-to-image model could learn to predict coherence for different views, mapping from coherence to image quality should be done for each group of views separately. Another advantage is that coherence can be used to give a global image quality metric without the need for a segmentation model. Finally, the pixel-based methods can be applied automatically in the most general way. Any classic metric can



(a) AI GLS[4] vs EchoPAC GLS (2D Strain tool)[5].



(b) AI EF[6] vs EchoPAC EF[7].

Figure 7. Box plots of the difference between clinical measurement values obtained automatically by AI [24,25] and reference measurements obtained manually using GE HealthCare EchoPAC on the HUNT4 data [13,23] per image quality category, as predicted by the end-to-end model. The decrease in standard deviation with better image quality indicates better agreement between the methods for higher image quality. Additionally, there is a noticeable change in bias between different quality categories. We believe this effect is partly caused by physiological differences correlated with image quality and is out of scope for this work. EF, ejection fraction; GLS, global longitudinal strain.

Table 8
Inference time comparison of the proposed methods in this work, implemented in C++ using the FAST framework [4]

Method	Inference time, ms
gCNR	3.45 ± 0.29
Local image coherence	5.65 ± 0.53
End-to-end model	1.53 ± 0.03

gCNR, generalized contrast-to-noise ratio.

Table 9
Inference time, parameter count and number of operations comparison of the neural networks used in this study

Model	Inference time in milliseconds	Parameters in millions	Multiply-accumulate (MAC) operations in billions
Segmentation nnU-Net	3.37 ± 0.27	33.5	16.65
Coherence prediction U-Net	2.20 ± 0.25	2.0	4.01
End-to-end network	1.53 ± 0.03	2.2	0.39

Inference time was tested on an NVIDIA A100 GPU using ONNX Runtime as inference engine. To measure the inference time, the model performed a warm-up run on 100 random dummy inputs. Afterward, the model performed inference on 1000 random inputs. The final inference time is the average runtime of each of the inference runs. These values do not include data-loading time and pre- and post-processing steps.

be applied on the ROI and background region given by the segmentation model. However, this approach also give the lowest accuracy compared to the other approaches.

The ablation study of the end-to-end learning model showed that increasing the complexity of the model did not improve performance. This is a result of the relatively small dataset and the specific task of the model. For a more general model of image quality prediction with more varied input, *e.g.*, one model for all cardiac views, a larger dataset and more complex model may be required.

Clinical use

Image quality estimation can be the first step toward a method for giving reliability estimates to clinical measurements and quality control of fully automatic methods. Figure 7 shows that variability in clinical measurements decreased with higher predicted quality. However, image quality is only one source of variability, so a reliability model would also need to include view correctness and other factors that determine whether a given input is difficult to assess.

More direct use cases of the quality prediction model include automatic selection of the best frame to perform a clinical measurement when multiple options are available, data cleansing in data mining and automatic disapproval of segments for regional strain analysis. All methods explored in this work are computationally efficient and can be run in real-time while scanning, and can thus be used as a guidance tool to enable clinicians to acquire images with better image quality.

Computational complexity

Table 8 compares the processing time of the three quality estimation methods described in this study. Runtimes were dominated by the inference times of the relevant neural networks, shown in Table 9. Both the gCNR method and local coherence method ran the nnU-Net segmentation model at inference. The coherence method additionally ran the coherence prediction model, assuming the coherence image was not available. The end-to-end method only ran the dedicated end-to-end model.

Real-time demonstration application and examples

To showcase the functionality of the end-to-end, real-time quality network, a real-time application was developed using the FAST framework [29]. The demo is a split-screen application that shows B-mode input to the left and segmentation regionally color-coded by quality as predicted by the end-to-end network to the right. Figure 9 provides a screenshot of the application in use. We also provide a demonstration video [30] illustrating the application in action while a clinician operates a GE Vivid E95 scanner, which can be accessed at https://doi.org/10.6084/m9.figshare.26413984.v2. This video demonstrates the output of the end-to-end neural network and how it reacts to different scenarios such as lung obstruction. The video also shows that the proposed end-to-end method can run in real-time while scanning and thus may be used to guide operators to achieve good image quality while scanning.

A video with more examples of image quality predictions on recordings from study participants in the HUNT4 study with high (BMI >30) and normal (20 < BMI < 25) BMI can be accessed at https://figshare.com/articles/media/Regional_quality_estimation_for_echocardiography_using_deep_learning_-_additional_examples/27730251?file=50487105.

Study participants with a high BMI had lower image quality on average. However, many other factors also influence image quality.

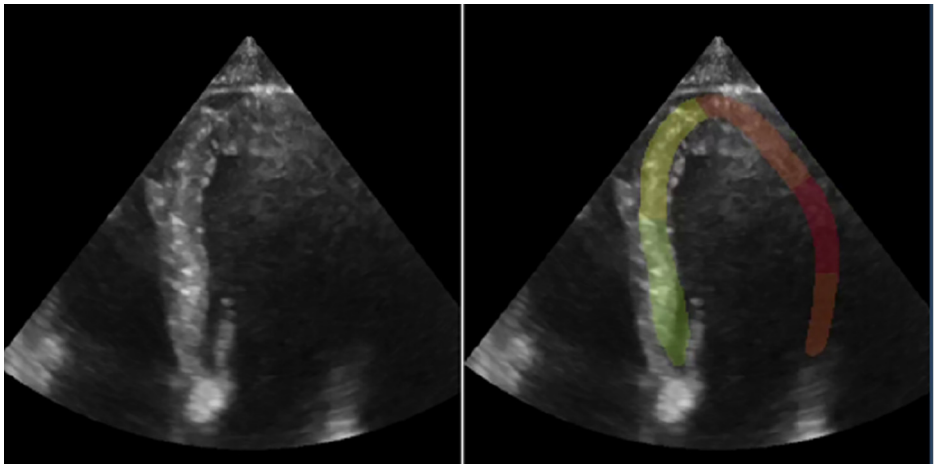


Figure 9. Screenshot of the real-time demo application. The left side shows the input B-mode image. The right side shows the output of the segmentation color-coded by the output of the end-to-end quality network. The color codes are the same as in Figure 5.

Conclusion

In this work, we developed and compared different deep-learning methods for regional image quality estimation in cardiac ultrasound. We showed that classic pixel-based methods, such as (g)CNR, together with automatic image segmentation, give low agreement with the quality assessment of cardiologists. We developed a U-Net model to predict the coherence factor for each pixel in the ultrasound image and showed that the resulting coherence image can be used to assess image quality in a pixel-based way with better performance than classical measures. The best results, below inter-observer variability, were obtained using an end-to-end deep-learning model. Finally, we showed that higher predicted quality is associated with lower limits of agreement between fully automatic and manual methods for the clinical measurements EF and GLS.

Conflict of interest

The authors declare no conflict of interest.

Acknowledgments

This work was supported in part by the Research Council of Norway under project no. 237887.

Data availability statement

The data used in this study is confidential and is therefore not publicly available.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Grammarly text editor and ChatGPT for assistance in sentence editing. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.ultrasmedbio.2024.12.008.

Appendix A. Extraction of cardiac regions of interest

We used nnU-Nett [31,32] architecture to segment the cardiac images. nnU-Net is used out of the box using the default configuration but without the final ensemble step. Instead, we trained and validated on a single predefined 80% train, 10% validation and 10% test split from the HUNT4 segmentation annotation dataset. Table A1 summarizes the characteristics of the nnU-Net architecture. This model is described in more detail and compared with other segmentation models in our previous work [33]. We used two nnU-Nets, one for apical two- (A2C) and four-chamber (A4C) views, and one for apical long-axis (ALAX) views. nnU-Net for A2C and A4C views segments the left ventricle (LV), left atrium (LA) and myocardium (MYO). nnU-Net for ALAX views additionally segments the aorta (AO).

Segmentation of the MYO was divided into eight regions using the following algorithm:

1. Extract the annulus points. For A2C/A4C views, these are the points where the MYO meets the LA. For ALAX views, these are the points where the MYO meets the LA and AO. Points A and B are the annulus points in Figure A1.
2. Extract the apex of the LV, defined as the furthest points from the base points within the LV lumen. This is point C in Figure A1.
3. Divide both the left and right part of the endocardium border, defined as the border between the LV and MYO regions, into three parts with equal length. This gives points D, E, F and G in Figure A1.
4. Find the closest points on the outer MYO border for points C, D, E, F and G. These are points H, I, J, K and L in Figure A1.
5. Fill in the regions by connecting the points via the contour, resulting in the MYO divided into six regions.
6. Draw circles (note: due to the unequal pixel spacing in depth and width, the annulus regions become ovals in the 256 × 256 segmentation maps. When plotting the images with equal spacing in width and depth, these regions become circles again) with a radius of 2 mm around the annulus points, points A and B in Figure A1. We used these additional two regions to assess the local image quality of the annulus points in the image. The result is the eight regions, as in Figure A1.
7. Remove any parts of regions that fall outside of the sector. The apical top regions in Figure 1a, *i.e.*, the yellow and white masks, are examples of this. If more than 50% of all pixels inside the region fall outside the sector, we exclude the region from analysis.

The goal is to automatically quantify the image quality in each of these eight regions. The LV lumen is used as background region.

Figure A1. Division of MYO into regions. A and B represent the MYO base points. C represents the LV apex. D, E, F and G are obtained by dividing the inner LV border into equal parts. H, I, J, K and L are obtained by finding the closest corresponding point on the outer MYO border.

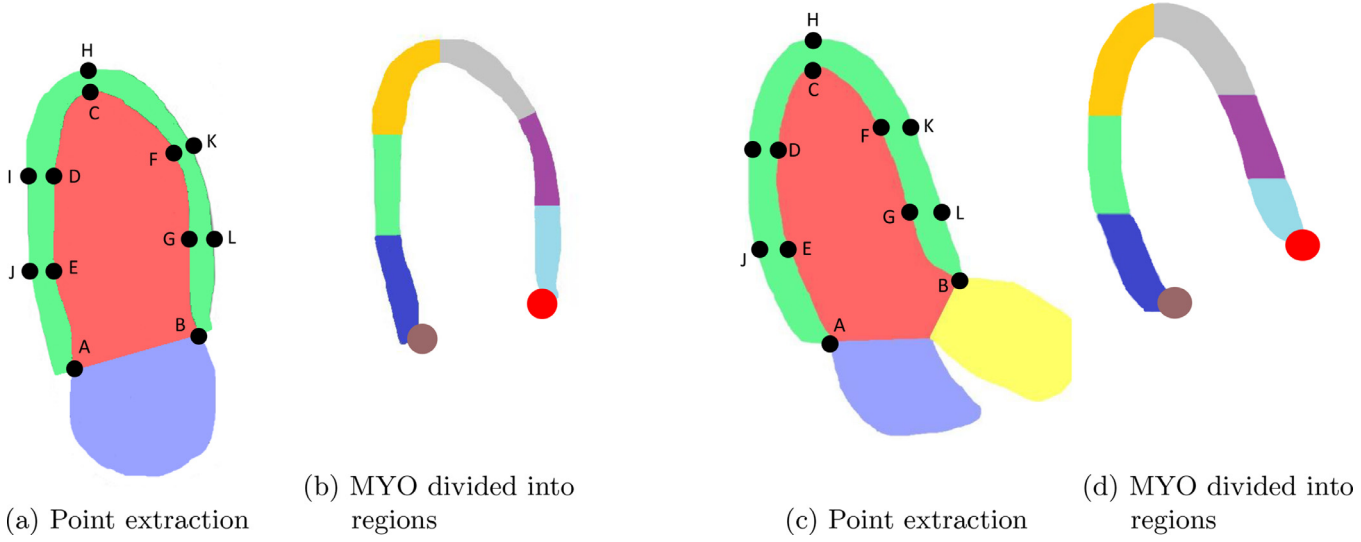


Table A1
Characteristics of the nnU-Net [5,6] used. The "number of channels" row indicates the number of channels at the first, bottom and last convolution of the U-Net, respectively. The number of output channels reflects the number of structures the model segments. For A2C/A4C views, the nnU-Net segments the LV, MYO and LA. For ALAX views, the nnU-Net additionally segments the AO.

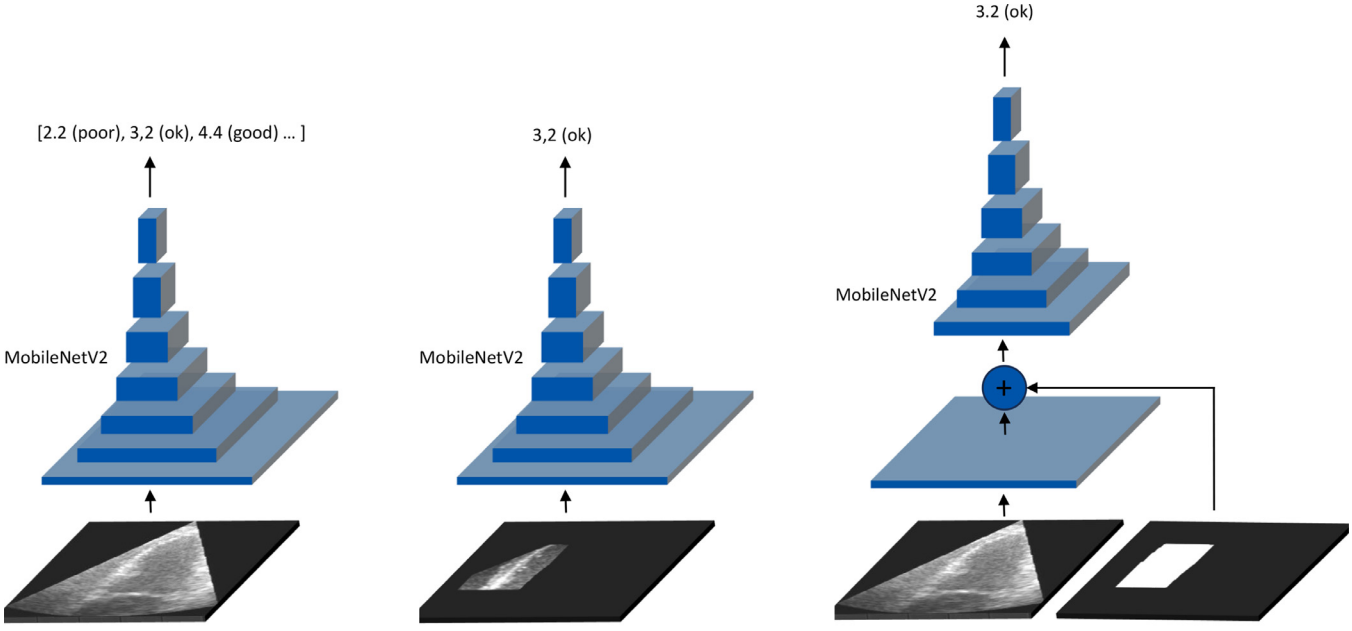
Input size	256 × 256
Number of channels	32 ↓ 512 ↑ 32
Number of output channels	3 for A2C/A4C views, 4 for ALAX views
Lowest resolution	4 × 4
Up-sampling scheme	Deconvolutions
Normalization scheme	InstanceNorm
Batch Size	49
Optimizer	Adam
Initial learning rate	1e-2
Scheduler	Polynomial
Loss function	Dice and cross-entropy
Inter-layer activation	Leaky Relu
Final layer Activation	Softmax
Epochs	500
Augmentations	Rotations, scaling, Gaussian noise, Gaussian blur, brightness, contrast, simulation of low resolution, gamma correction and mirroring. For more details, see Isensee et al. [6]

Appendix B. Ablation study of the end-to-end learning model

In the ablation study, we evaluated the impact of modifying the architecture of the end-to-end learning model. Three different network architectures were tested: Cardiac View Classification (CVC) network [3], MobileNetV2 [20] and EfficientNet [34]. We tested approaching the problem both as a classification and regression task, with only the final dense layer and loss function being changed accordingly. Additionally, three basic network attention variations were tested using the automatic segmentation output. The default model did not use any attention and predicted each label directly, as in Figure B1a. For the other two versions, the region masks extracted from the segmentation were dilated with a square dilation filter of 50 × 50 pixels and used as an additional input to the networks that then predicted the label one region at a time. The dilation filter revealed the direct vicinity around each region so the boundary between tissue and background became visible. The first variant used this dilated mask as hard attention by blacking out the other parts of the image, as shown in Figure B1b. For the second variant, the masks were used as soft attention as input to a side branch of the network, as proposed by Eppel [35]. For this version, we created an attention map and added it element-wise to the output of the first layer, corresponding to version "c" in Eppel [35]. Figure B1c shows this configuration.

The ablation study consisted of two parts. Both parts used the training configuration listed in Table 4 and data from the regional image quality dataset. In the first part, we examined the effect of changing the convolutional backbone and the effect of framing the problem as a classification or regression task. Table B1 compares the predictions with annotations on the test set for the different configurations using the default end-to-end model. In the second part of the ablation study, the backbone was fixed to MobileNetV2 [20] and the problem was framed as a regression task. Next, the different variants shown in Figure B1 were compared. Table B2 summarizes the results on the test set. The default end-to-end model with no attention, the MobileNetV2 [20] architecture and the problem framed as a regression task gave the best results and were thus used for this paper.

Figure B1. Different variants of the end-to-end learning model. The default model outputs a list of all quality scores at once, while the other two variants output only a single quality score. The default model (a) is a plain MobileNetV2 that predicts the quality labels of each region at once. The hard attention model (b) predicts the quality of each region one at a time. The input to the network is a masked version of the ultrasound image where all pixels that do not belong to the region and its immediate surroundings are blacked out. The soft attention model (c) also predicts the quality of each region one at a time. Here the input consists of two images, the ultrasound image and an attention mask highlighting the region and its immediate surroundings. The end-to-end soft attention model corresponds to version c in Eppel [34].



(a) Default end-to-end model (b) End-to-end hard attention model (c) End-to-end soft attention model

Table B1
Results on the test set for the first part of the end-to-end learning ablation study

Task	Model [Ref.]	Spearman correlation	Mean absolute error	Number of parameters, millions
Regression	CVC [7]	0.62	0.60 ± 0.45	9.6
	MobileNetV2 [9]	0.65	0.59 ± 0.45	2.2
	EfficientNet-B0 [10]	0.63	0.61 ± 0.45	4.0
	EfficientNet-B4 [10]	0.63	0.60 ± 0.44	18
	EfficientNet-B7 [10]	0.63	0.59 ± 0.46	64
Classification	CVC [7]	0.38	0.91 ± 0.81	9.7
	MobileNetV2 [9]	0.48	0.80 ± 0.71	2.3
	EfficientNet-B0 [10]	0.46	0.80 ± 0.76	4.1
	EfficientNet-B4 [10]	0.46	0.83 ± 0.73	18
	EfficientNet-B7 [10]	0.37	0.91± 0.79	64

The CVC [7] network is trained from scratch. The other networks are pre-trained on ImageNet [8]. Each method is compared against annotations from the test set of the regional image quality dataset. CVC, Cardiac View Classification.

Table B2
Results on the test set for the second part of the end-to-end learning ablation study using Mo-bileNetV2

Model variant	Spearman correlation	Mean absolute error
Default (no attention)	0.65	0.59 ± 0.45
Variant 1 (hard attention)	0.47	0.68 ± 0.60
Variant 2 (soft attention)	0.53	0.67 ± 0.57

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